MAS · Statistical Inference (Sample Mean) — Patterns and Solution Strategies

Based on: 17 past exam questions from 2016–2025, marking keys + exam reports. Appears only in Calculator-assumed papers. Original questions in `topics/07_Statistical_Inference.pdf`. Numerical examples verified with scipy. Also references 2016 sample papers A9, A13 (original papers in `papers/2016Sample_*`). Review status: has undergone one round of independent review and been revised according to feedback (see `Review_log.md`).

Formula card

- Sample mean: $\bar{X} \sim N(\mu, \sigma^2/n)$ (approximately). If the population itself is normal, this is exact for any n; if the population is not normal, by the Central Limit Theorem it is approximately true when n is sufficiently large (in practice ≥ 30), regardless of the shape of the population.
- Standard deviation of the sample mean (standard error): $\sigma_{\bar{X}} = \sigma/\sqrt{n}$.
- Sum of n variables: the key's method is to replace "sum < T" with "$\bar{X}$ < T/n" (2016A-Q19(b), 2020A-Q18(a)). Sum ~ $N(n\mu, n\sigma^2)$ is only for checking; the key does not give this form.
- Confidence interval for μ: $\bar{x} \pm z\cdot\sigma/\sqrt{n}$ (when σ is unknown, use the sample standard deviation s instead); z = 1.645 / 1.960 / 2.576.
- Working backwards from an interval: $\bar{x}$ = interval midpoint; $z\cdot\sigma/\sqrt{n}$ = half-width ⇒ can solve for $\sigma/\sqrt{n}$, s or n.

Marking-key reminders

- Write mathematical statements, not calculator language (2016, 2017, 2018 reports): write $P(\bar{X} > 83) = P(Z > 1.5) = 0.0668$, not normCdf(83, ∞, 80, 2).
- "State the distribution" 3 marks: ✓ state that $\bar{X}$ is approximately normal (writing n > 30 or CLT as well is safer; each year's key only requires "normal" for the 1st mark) ✓ mean ✓ standard deviation $\sigma/\sqrt{n}$ (or variance).
- When explaining, make clear which quantity is meant: "the standard deviation of the sample mean decreases", not just "standard deviation decreases", and do not use "it" (2019 report).
- Answer the question asked; do not copy sentences from previous years' standard answers (2025 report).

Pattern 1: Write down the distribution of $\bar{X}$ and find the probability ★ Frequently tested

Standard procedure

1. "By the Central Limit Theorem, since n = 100 is large, $\bar{X}$ is approximately normal with mean μ = 80 and standard deviation $\sigma/\sqrt{n} = 20/10 = 2$."
2. Write the probability statement, then calculate it.
3. "Within d of μ" = $P(|\bar{X} - \mu| < d) = P(\mu - d < \bar{X} < \mu + d)$.
4. Sum type: total over 50 times < 8.96 kL ⇔ $\bar{X}$ < 179.2 L ⇒ $\bar{X} \sim N(175, 2.1213^2)$ ⇒ P ≈ 0.9761 (2016A-Q19(b)).

When the population is not normal (uniform distribution 2017A-Q9, exponential distribution 2018A-Q12, right-skewed pdf 2021A-Q15, dartboard distance 2025A-Q13): the method is the same as for a normal population, because n is sufficiently large, so $\bar{X}$ is still approximately normal. 2018A-Q12(c) specifically tests this point (2 marks: answer unchanged + reason is the CLT). Drawing the distribution of $\bar{X}$: a bell shape centred at μ, with width about $\pm 3\sigma/\sqrt{n}$, much narrower than the original distribution; when the question requires it, mark the probability region calculated in the previous part on the graph (2017A-Q9(b) has 1 mark, 2021A-Q15(b)).

Past questions: 2016A-Q19(a)(b), 2017A-Q9, 2018A-Q12(a)(b), 2019A-Q14(a)(b), 2020A-Q18(a), 2021A-

Q15(a), 2022A-Q12(a)(b), 2023A-Q13(a)(b), 2024A-Q14(a)(b), 2025A-Q13(a)(b).

Pattern 2: Find the sample size n (so that a probability or interval meets requirements)

Standard procedure

1. First draw a diagram to decide whether it is two-sided, one-tailed or a half-interval, then write the probability statement. Two-sided: $P(|\bar{X} - \mu| < 0.2) = 0.98$; one-tailed: $P(\bar{X} > 25) = 0.03 \Rightarrow z = 1.881$ (2018A-Q12(d)); half-interval: $P(80 < \bar{X} < 82) > 0.4 \Rightarrow P(0 < Z < k) = 0.4$, $k = 1.282$, $n = 165$ (2019A-Q14(d)).
2. Find the critical z (98% ⇒ z = 2.326).
3. Set up an equation or inequality: $z \cdot \sigma/\sqrt{n} = d \Rightarrow n = (z\sigma/d)^2$.
4. Rounding: when "at least" is required, round up. Example: $(2.326 \times 1.5/0.2)^2 = 304.3 \Rightarrow n = 305$ (2020A-Q18(b)); 600.25 ⇒ 601 (2024A-Q14(f), when "at least" is required, round up no matter how small the decimal part is). When working backwards from a known interval to find n, take the nearest integer (2022A-Q14(a): using z = 2.576 gives 200.0 ⇒ 200; the key also accepts using 2.58 to get 201; 2025A-Q13(c): 283.0 ⇒ 283, with 284 also accepted).

Past questions: 2016A-Q19(d), 2018A-Q12(d), 2019A-Q14(d), 2020A-Q18(b), 2022A-Q14(a), 2024A-Q14(f), 2025A-Q13(c); 2016S-A9, 2016S-A13(b) (sample papers). Working backwards to find the margin of error: 2024A-Q14(c) (50% probability corresponds to m ≈ 1.7 minutes).

Pattern 3: Calculating confidence intervals and working backwards

Calculation: $\bar{x} \pm z \cdot s/\sqrt{n}$. Example: $\bar{x} = 10300$, s = 400, n = 100, 95% ⇒ $10300 \pm 1.96 \times 40 = (10221.6, 10378.4)$ (2017A-Q13(a): 1 mark each for the centre, standard error, z, and lower/upper limits). Working backwards

- Upper limit 40.62, width 1.08 ⇒ $\bar{x} = 40.08$; $0.54 = 2.576 \times s/20 \Rightarrow s \approx 4.19$ (2018A-Q16).
- Interval $150 \le \mu \le 200 \Rightarrow \bar{x} = 175$, $\sigma_{\bar{X}} = 25/1.96 \approx 12.76$; after doubling the sample size, $\sigma_{\bar{X}} = 12.76/\sqrt{2} \approx 9.02 \Rightarrow P(|\bar{X} - \mu| < 10) \approx 0.7325$ (2020A-Q17).
- Given an interval and n, work backwards to find the confidence level: first find z = half-width / standard error, then look up the corresponding percentage (2017A-Q13(d)).
- Change in sample size: width $\propto 1/\sqrt{n} \Rightarrow$ to make the width 1/4, 16n is needed (2021A-Q17(c)); if the sample size becomes 2n, the standard error is divided by $\sqrt{2}$.

Past questions: 2016S-A13, 2017A-Q13, 2018A-Q16, 2019A-Q15, 2020A-Q17, 2021A-Q17, 2022A-Q12(c), 2023A-Q13(c)(d), 2023A-Q15, 2024A-Q14(d).

Pattern 4: Interpreting confidence intervals (true/false) ★ Examined almost every year (except 2016, 2020)

Correct statements to memorise

1. μ is a fixed but unknown number. A particular interval either contains μ or does not, and there is no way to know which (2018A-Q16(d), 2019A-Q15(d), 2021A-Q17(e), 2022A-Q14(c)).
2. "95% confidence" means that if sampling were repeated, about 95% of intervals would contain μ, not that "the probability μ lies in this interval is 95%" (2023A-Q15(c): Ali's statement is incorrect).
3. A wider interval does not mean it is more likely to contain μ (2022A-Q14(c): James is incorrect).
4. The mean of the next sample does not necessarily lie in this interval (2017A-Q13(b)(i)); an individual value is even less likely to, because the variation between individuals is much greater than the variation of the sample mean (2017A-Q13(b)(ii)).
5. If 10 of 50 intervals have lower limits below 1.00, this does not mean the setup is wrong; it is normal variation from random sampling (2023A-Q13(e)). Conversely: if, among a large number of intervals, the proportion that do not contain μ is far above 5%, that is evidence. In 2025A-Q13(e), 298 of 500 95% intervals do not contain μ = 100; the key judges Chen's statement 1 (the sample is unlikely to

come from a population with μ = 100) to be correct; but the reason still cannot be determined, so statement 2 (the competitor did not stand at 3.50 m) is incorrect.

6. If two intervals are nested or overlap, this does not imply that the two population means are the same; nor can you say the new method is better just because its sample mean is lower (2024A-Q14(e); the key's reason for Sanjeet is: the two intervals are based on different sample sizes, so they cannot be compared directly in this way).
7. Duplicating the same sample to make 2n is not a new random sample, does not reflect real variation, and the interval does not actually become narrower (2022A-Q12(d)). Note: combining two independent random samples is valid, gives a larger n, and gives greater precision (2019A-Q15(c)).

How to write it: first state your position (True / False / Not correct), then give one reason in the context of the question; 1 mark each.

Pattern 5: Using an interval to test a claim (does a company use more water?)

Standard procedure (4 marks)

1. From the new sample, find $\bar{x}$ and s: 6.57 kL / 36 = 182.5 L, s = 15.
2. Construct a confidence interval for μ_new: 182.5 ± 1.96 × 15/6 = (177.6, 187.4).
3. Compare: the original μ = 175 is not in the interval.
4. Conclusion: there is evidence to support "WolliWorks uses significantly more water" (2016A-Q19(e)); conversely, if it were in the interval, there would not be enough evidence (2020A-Q18(c)).

Note: you can only draw a conclusion about "whether the means are different"; you cannot infer the source of the sample (for example, Anika's claim that it "probably came from 100 teenagers"), because the source is unknown (2021A-Q15(c)).

Past questions: 2016A-Q19(e), 2020A-Q18(c), 2021A-Q15(c), 2025A-Q13(d)(e).

Pattern 6: Factors affecting precision

Change	Standard deviation of $\bar{X}$	Interval width	Tail probability such as "$\bar{X}$ > 83" (2019A-Q14: μ = 80, threshold 83 is above μ)
n increases	decreases	narrows	decreases (83 is relatively more extreme)
Confidence level increases	—	widens	—
σ (or s) increases	increases	widens	increases

The larger the sample and the smaller the standard error, the higher the precision (2019A-Q15(c): combined sample 3n has the highest precision). To compare which of two intervals is narrower: compare $z \cdot s/\sqrt{n}$ (2021A-Q17(f)).